SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-020: Student Self-Service Password Change

Field	Details
Change Request ID	CR-020
Change Request Title	Student Self-Service Password Change
Student Name	J.Thushany
Student ID	JF24ICT09
Implementation Status	Completed
Date	13 October 2026
Project	Student Management System
Baseline Version	Student Management System v1.0
Environment	Development

1. Change Request Summary

1.1 Business Requirement

Students need a secure and convenient way to change their account passwords without requiring administrator assistance. The change request introduces a password change facility within the Student Dashboard. Students can enter their current password, provide a new password, and re-enter the new password for confirmation.

The feature must validate the submitted information, securely store the new password using password hashing, and display suitable success or error notifications.

1.2 Existing Behaviour

Before implementation:

- Students could log in using their existing credentials.
- Students could access their personal profile through the Student Dashboard.
- A dedicated student self-service password change facility was not available.
- Students did not have a dedicated interface to verify their current password and update their own password.

- Password updates required an additional implementation to support student self-service functionality.

1.3 Requested Change

The following functionality was required:

- Add a Change Password section to the Student Dashboard.
- Provide fields for Current Password, Password, and Re-enter Password.
- Validate that all required fields are completed.
- Verify that the new password and re-entered password match.
- Verify the student's current password before updating it.
- Enforce the existing password complexity requirements.
- Hash the new password using bcrypt before storing it in the database.
- Allow an authenticated student to change only their own password.
- Display appropriate success and error notifications.
- Provide a Clear button to reset the form fields.
- Return focus to the new password field after clearing the form.
- Preserve existing student profile and login functionality.

2. Implementation

2.1 Implementation Summary

CR-020 was implemented by adding a student self-service password change facility to the Student Dashboard.

A new Change Password component provides the password input fields, client-side validation, submission handling, notifications, and form clearing functionality. The component sends a password change request to the backend API.

The backend controller validates the submitted values, checks that the current password matches the stored password hash, verifies that the new password satisfies the password requirements, and updates the password hash using bcrypt.

The password change endpoint is protected by authentication and student-role authorization. The authenticated user's identifier is used to identify the account being updated.

2.2 Affected Files

File	Change	Purpose
backend/src/models/studentModel.js	Modified	Added model functions to retrieve and update the password hash associated with the authenticated user account.
backend/src/controllers/studentController.js	Modified	Implemented password validation, current-password verification, password hashing, and password update handling.
backend/src/routes/studentRoutes.js	Modified	Added the authenticated student password change endpoint while preserving existing student routes.
frontend/src/components/ChangeStudentPassword.jsx	Added	Implemented the password change form, validation, notifications, and Clear functionality.
frontend/src/pages/StudentDashboardPage.jsx	Modified	Integrated the Change Password component into the Student Dashboard.
frontend/src/styles/studentDashboard.css	Modified	Added styling for the password form, action buttons, and notification messages.

2.3 Technical Changes

Password Change Interface

The Student Dashboard now includes a Change Password section with three fields:

- Current Password
- Password
- Re-enter Password

The form includes Change Password and Clear buttons.

Input Validation

The implementation checks that all required password fields contain values. It also verifies that the new password and confirmation password match.

The existing password validation rules require at least eight characters, including uppercase and lowercase letters, a number, and a special character.

Current Password Verification

The backend compares the supplied current password with the stored password hash using bcrypt. If the current password is incorrect, the password update is rejected.

Secure Password Storage

The new password is hashed using bcrypt before the database is updated. The plaintext new password is not stored as the account password.

Authentication and Authorization

The password change endpoint requires an authenticated student session. The backend uses the authenticated user's identifier to update the corresponding account, rather than accepting another user's identifier from the form.

Form Clearing and Notifications

The Clear button resets the password fields and returns focus to the new password field. Success and error notifications provide feedback about the result of the operation.

2.4 Implementation Logic

1. The student signs in and opens the Student Dashboard.
2. The Change Password component is displayed below the student profile.
3. The student enters the current password.
4. The student enters a new password and re-enters it for confirmation.

5. The frontend checks that all fields are completed and the new passwords match.
6. The frontend sends a PATCH request to /api/students/change-password.
7. The backend authenticates the request and verifies the student's role.
8. The backend checks the current password against the stored password hash.
9. The backend validates and hashes the new password.
10. The password hash is updated for the authenticated user's account.
11. The frontend displays a success or error notification.
12. The student can log out and attempt to sign in using the new password.

2.5 Error and Data Handling

The implementation handles the following situations:

- Missing password fields.
- New password and confirmation password mismatch.
- Incorrect current password.
- New password that does not satisfy the password requirements.
- Unauthenticated requests.
- Requests from users without the required student role.
- Password update failures.

The backend is responsible for enforcing security-related validation. Frontend validation alone is not relied upon to authorize password changes.

3. Testing and Verification

3.1 Testing Approach

Manual browser testing was performed to verify the password change workflow.

Testing included:

- Functional testing.
- Input validation testing.
- Password confirmation testing.

- Incorrect current-password testing.
- Successful password update testing.
- Login regression testing.
- User interface testing.

Field	Details
Testing Environment	Development
Browser	Google Chrome
Test Role	Student
Test Method	Manual browser testing
Test Status	All five reported tests completed successfully

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Submit the form with empty password fields.	The form should reject the submission and display a validation message.	The required-field validation was checked in the browser.	Pass
TC-02	Enter different values in Password and Re-enter Password.	The system should reject the submission and display a password mismatch message.	The password mismatch validation was checked.	Pass
TC-03	Enter an incorrect current password with matching new password fields.	The backend should reject the password change.	The incorrect-current-password test was completed.	Pass
TC-04	Enter the correct current password and a valid matching new password.	The password should be updated and a success notification should be displayed.	The password change was completed successfully.	Pass

TC-05	Log out and sign in using the new password.	Login should succeed with the new password.	Login using the new password was completed successfully.	Pass

3.3 Test Summary

Test Result	**Count**
Total Test Cases	5
Passed	5
Failed	0
Pass Rate	100%
Overall Test Result	Pass

Note: The results above reflect the five browser tests reported as completed. The screenshots attached to this report should correspond to the actual test results.

4. Acceptance Criteria Verification

ID	**Acceptance Criterion**	**Verification Method**	**Status**
AC-001	The Change Password section is available on the Student Dashboard.	Browser UI verification	Met
AC-002	The form provides current password, new password, and confirmation fields.	Form inspection	Met
AC-003	Empty fields and mismatched passwords are rejected.	Validation testing	Met
AC-004	The current password is verified before the update.	Incorrect current-password test	Met
AC-005	The new password satisfies the password complexity requirements.	Password validation	Met
AC-006	The new password is securely hashed before database storage.	Backend implementation review	Met

AC-007	Only the authenticated student can update their own password.	Authentication and authorization implementation review	Met
AC-008	Success and error notifications are displayed.	Browser testing	Met
AC-009	The Clear button resets the fields and returns focus to the password field.	UI interaction testing	Met
AC-010	The student can log in using the new password after changing it.	Login regression test	Met

5. Evidence

Five screenshots are included to document the browser testing performed for CR-020. This provides evidence beyond the minimum two screenshots requested.

Evidence 01 – Change Password Interface

Related criteria: AC-001, AC-002.

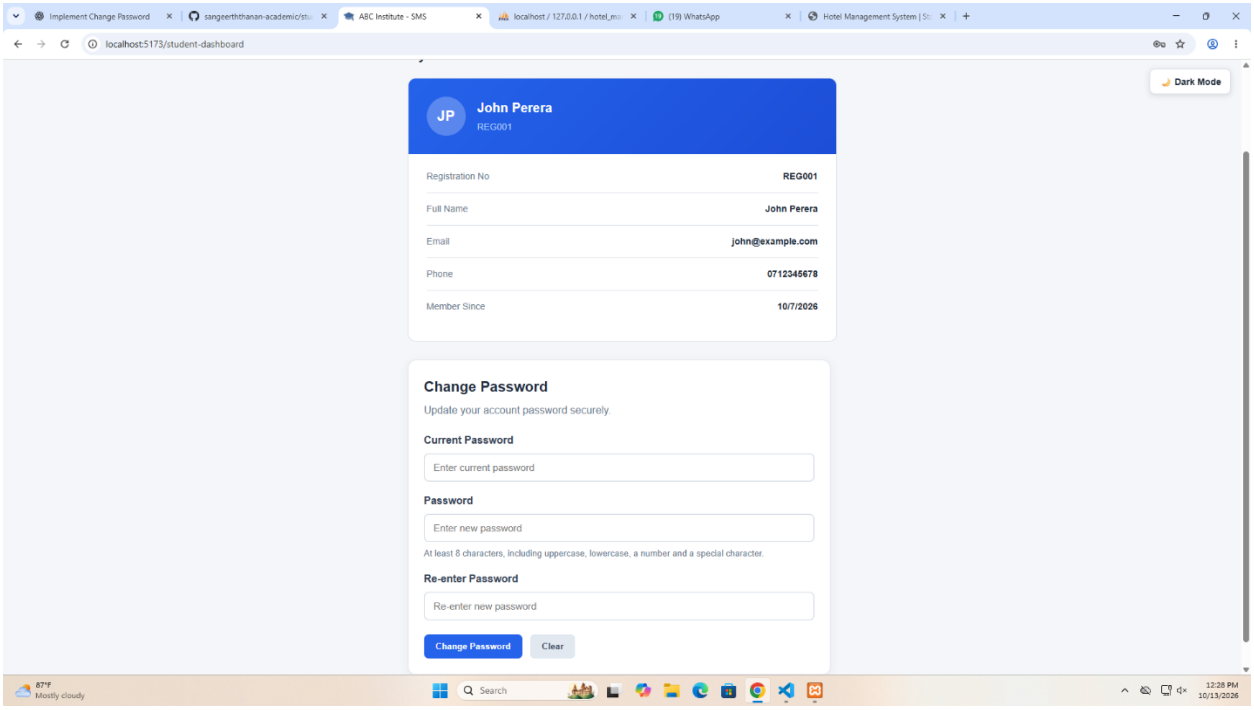


Figure 1: Student Dashboard displaying the Change Password section and its three password input fields.

Evidence 02 – Password Confirmation Validation

Related criteria: AC-003.

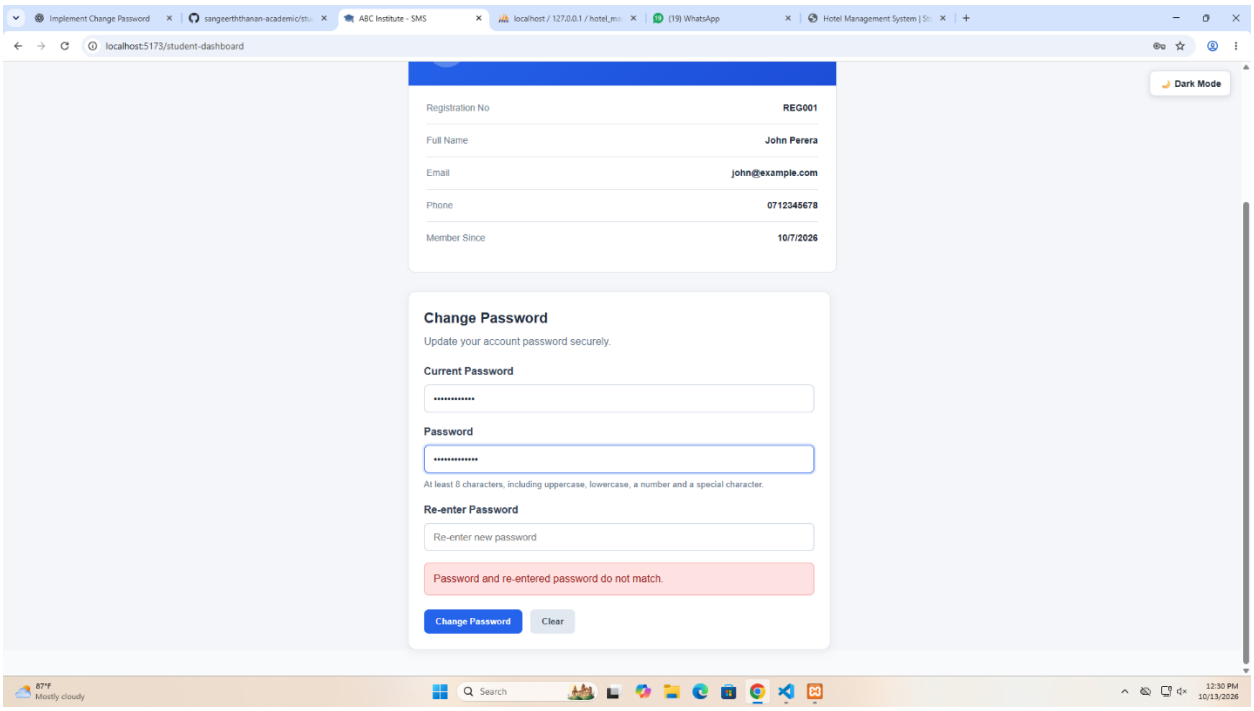


Figure 2: Change Password form displaying the validation message when the new password and confirmation password do not match.

Evidence 03 – Incorrect Current Password

Related criteria: AC-004.

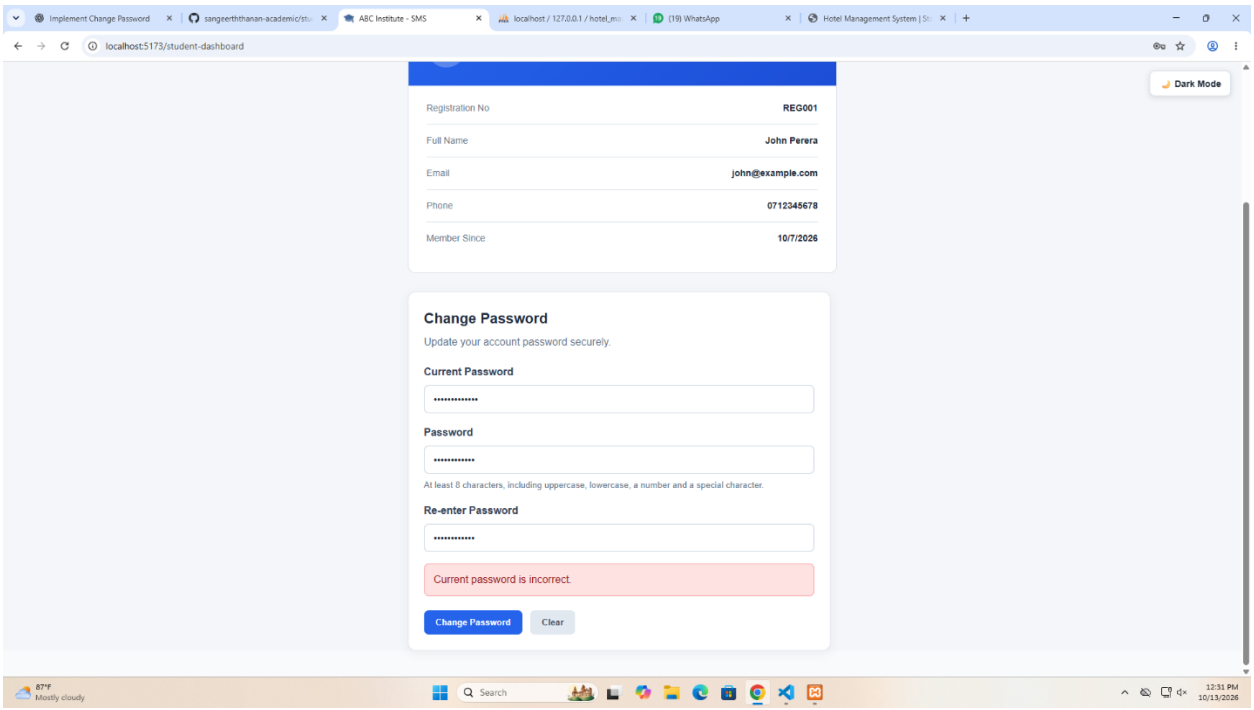


Figure 3: Change Password form displaying the error message after an incorrect current password is submitted.

Evidence 04 – Successful Password Change

Related criteria: AC-005, AC-006, AC-008.

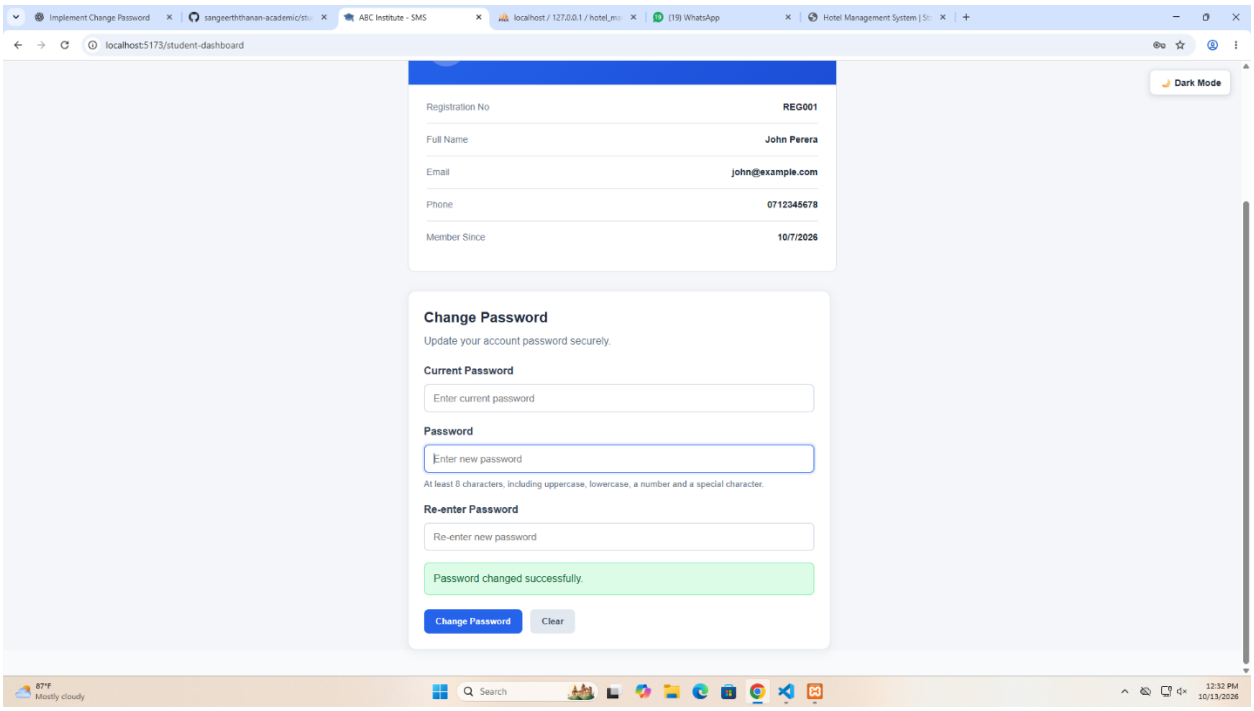


Figure 4: Change Password form displaying the success notification after a valid password change.

Evidence 05 – Login Using the New Password

Related criteria: AC-010.

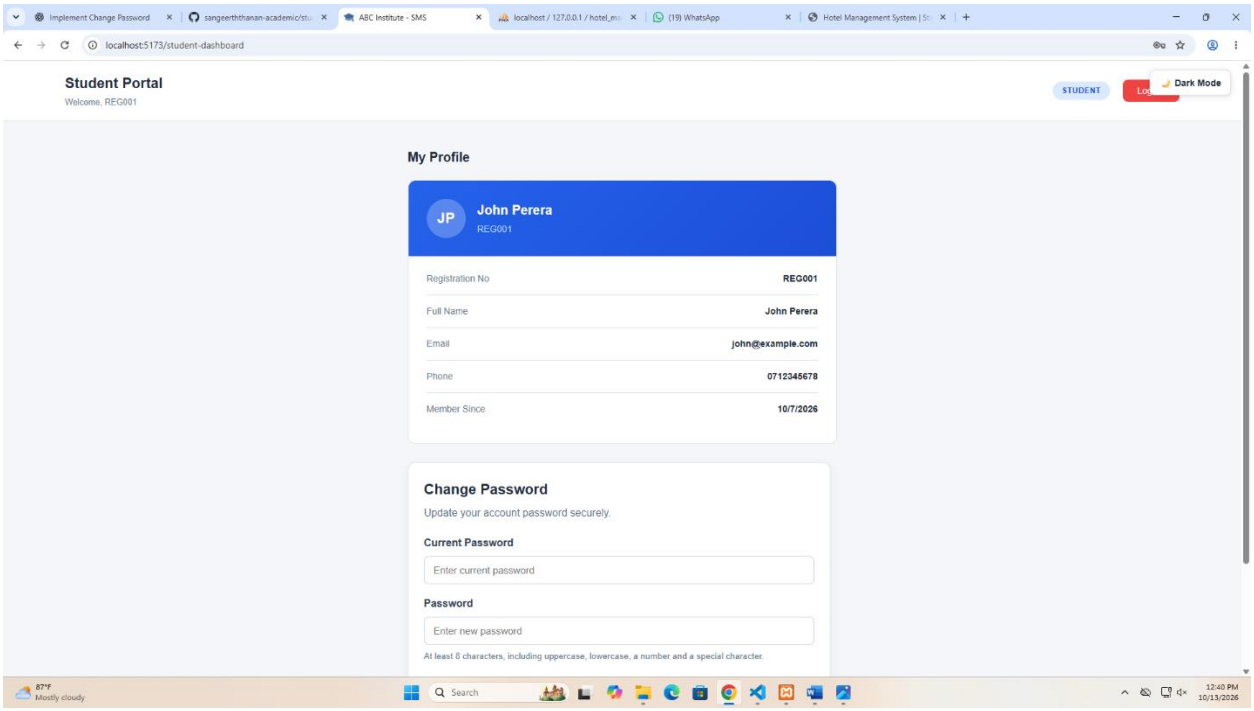


Figure 5: Student successfully signing in using the new password after logging out of the previous session.

6. Final Result

Item	Result
Requirements	Met
Implementation	Completed
Testing	Passed
Acceptance Criteria	Met
Existing Profile Functionality	Preserved
Existing Login Functionality	Verified through new-password login
Final Status	Completed

6.1 Final Summary

CR-020 was implemented to provide students with a self-service password change facility within the Student Dashboard. The implementation includes current-password verification, new-password confirmation, password complexity validation, bcrypt password hashing, authentication and role protection, form clearing, and success or error notifications.

The password change interface communicates with the backend through the /api/students/change-password endpoint. The backend validates the request and updates the password hash associated with the authenticated user's account.

Five manual browser tests were reported as completed, covering required-field validation, password mismatch, incorrect current password, successful password change, and login using the new password. All five tests were reported as passed.

6.2 Issues / Limitations

No outstanding issues were reported during the five browser tests. The screenshots should be attached to provide supporting evidence for the implementation and testing results.

Student Name: J.Thushany

Student ID: JF24ICT09

Date: 13 October 2026

Signature: _____

~ END OF REPORT ~